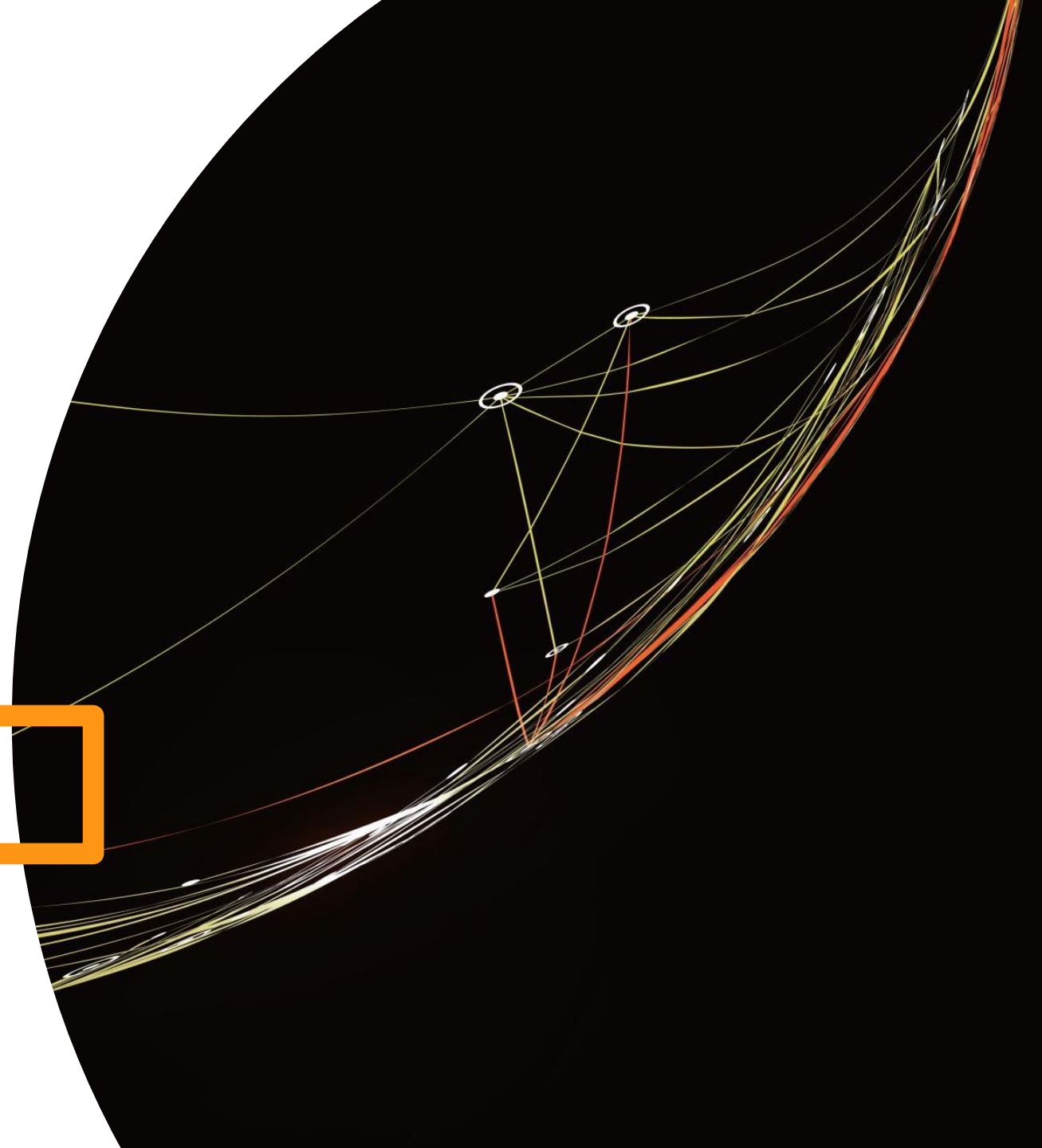




SHA-512

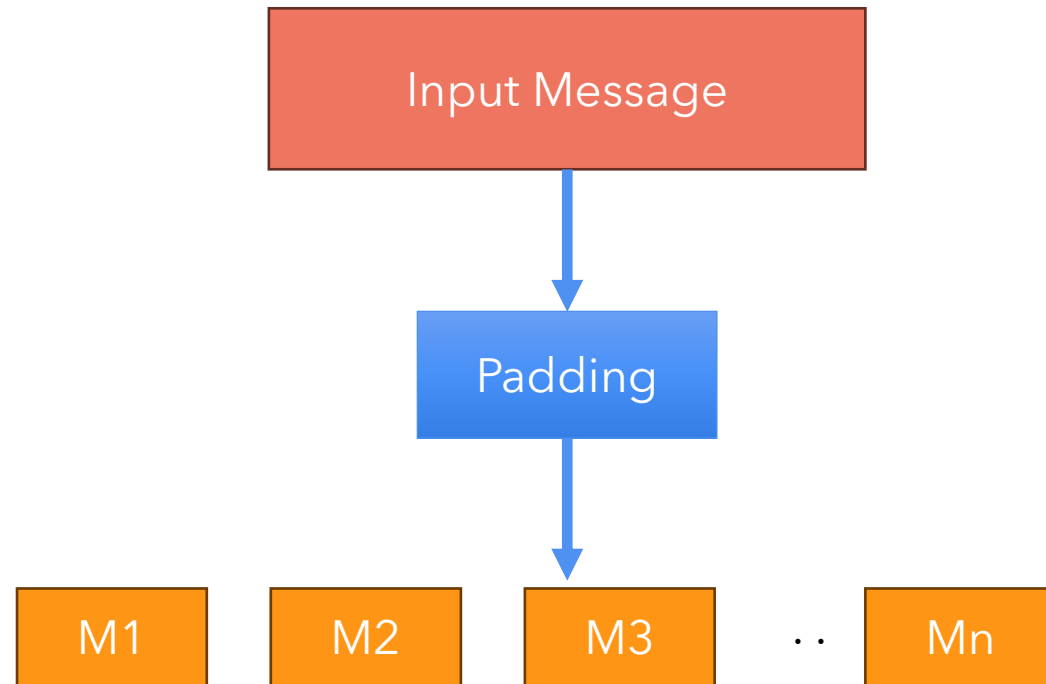
Module IV



SHA-512 secure hash Algorithms

- **Introduction to SHA:**

SHA0, SHA-1 (160 bits), SHA-2 (SHA-224, SHA-256, SHA-384, SHA-512).



Padding

• abc => 01100001 01100010 01100011

• Message length (M) = 24 bits.

Needed,

• Message length (M) $24 \equiv 896 \pmod{1024}$

• $24 \pmod{1024} = 896$

• $24 \neq 896$

• $24 + 872 \pmod{1024}$

• $896 = 896 \checkmark$

• Pad 872 bits to message such that $M \pmod{1024} = 896$

• 872 bits to be padded $\rightarrow$ 1 bit followed by 871 zeros.

Padding

```
6162638000000000 0000000000000000 0000000000000000 0000000000000000
0000000000000000 0000000000000000 0000000000000000 0000000000000000
0000000000000000 0000000000000000 0000000000000000 0000000000000000
0000000000000000 0000000000000000 [ 896 bits + 128 bits remaining ]
```

Pad the original length of the message for 128 bits at the end.

Message length = 24 bits

Hexadecimal value of 24_{10} is 18_{16} .

Message length represented in 128 bits in hexadecimal is 0000 0000 0000 0000
0000 0000 0000 0018.

Padding- example

2348 bits $\Rightarrow 1024 \times 2 = 2048$.

Message length (l) = $896 \bmod 1024$.

$2348 \bmod 1024 = 300 + 596$ (896)

$2348 + 596 \bmod 1024 = 896$

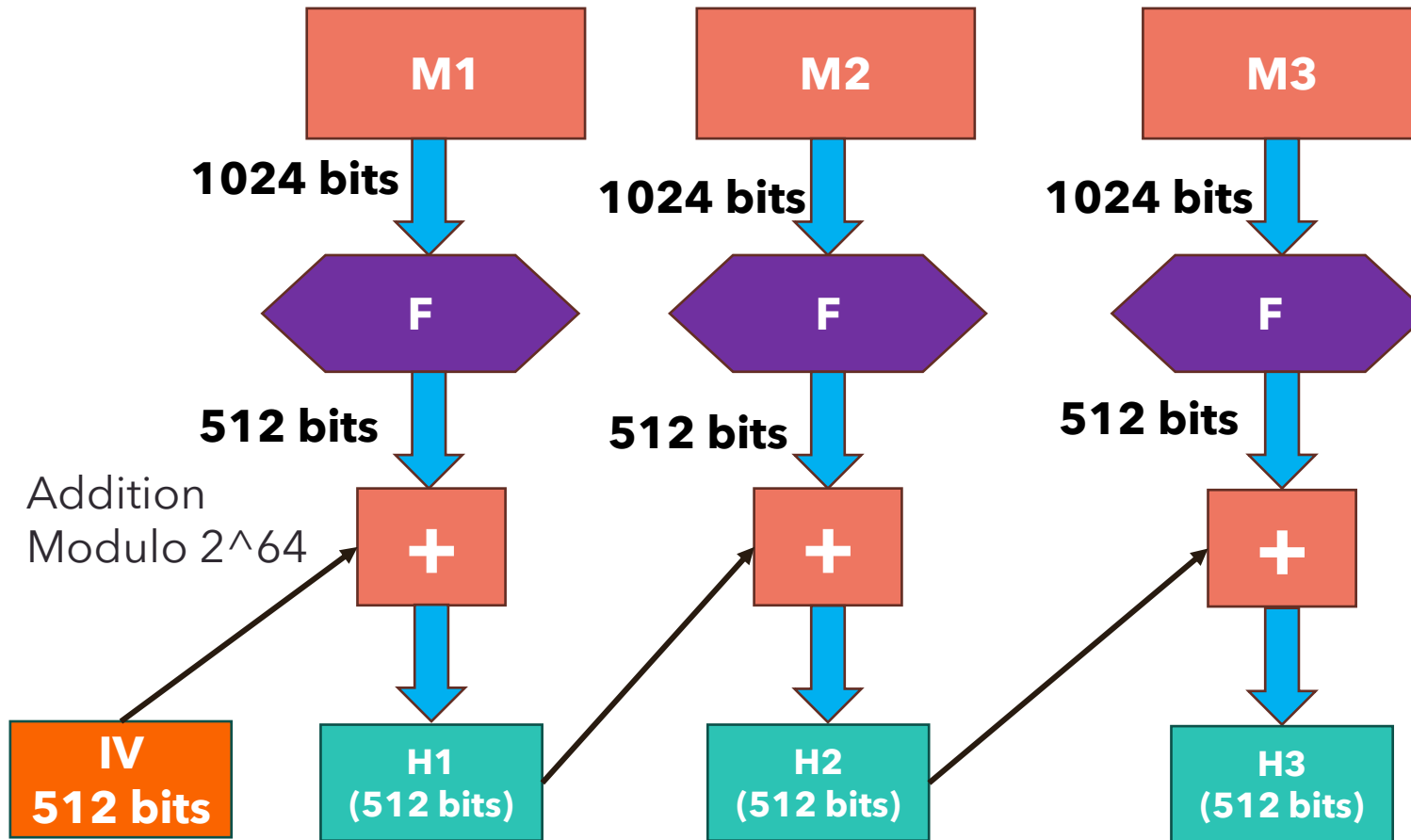
pad (1 || 595 zeros)

Message length l = $2348 + 596 = 2944$.

Add the message length (2348) as 128 bits at the end.

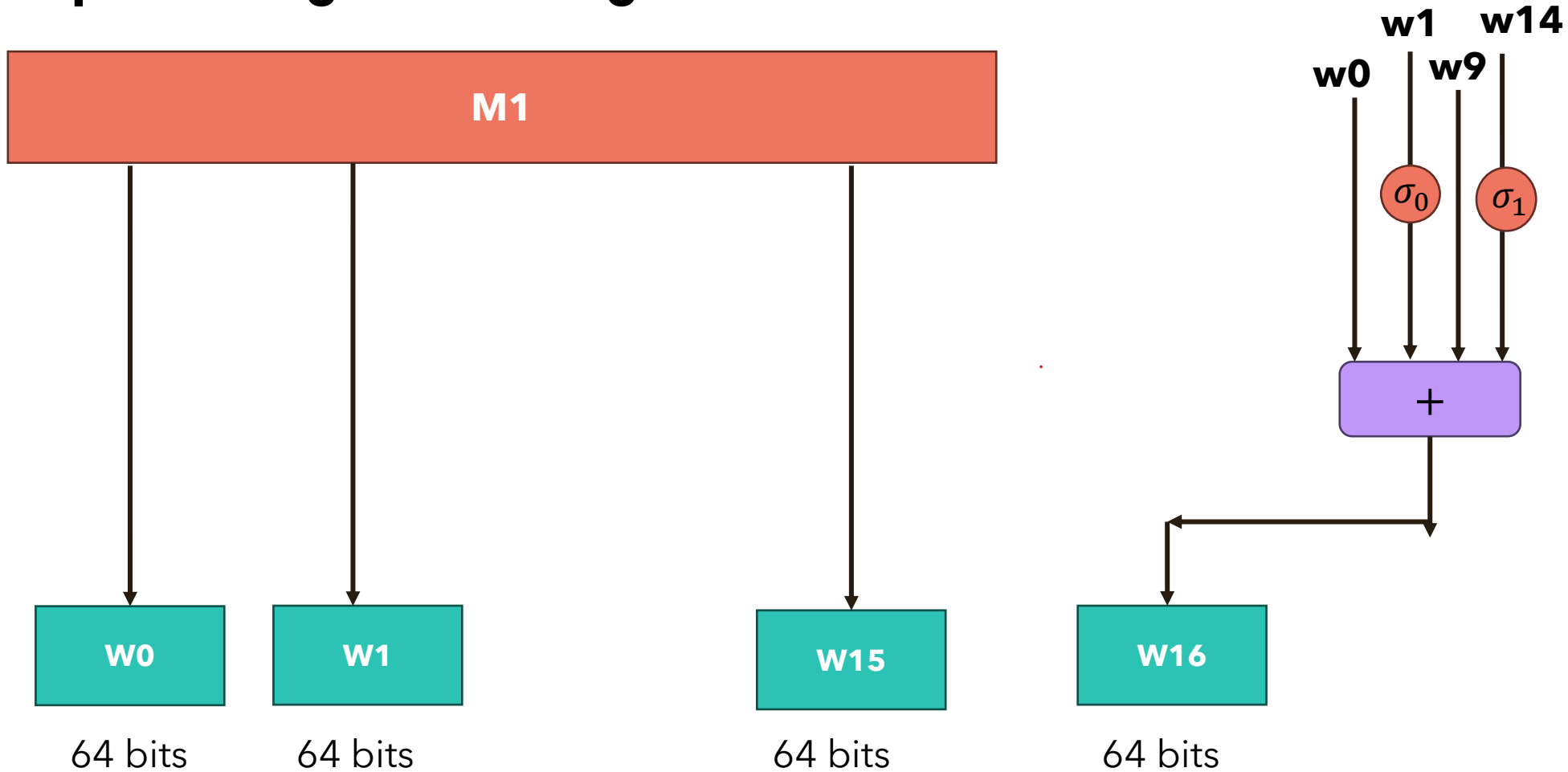
Total bits = $2944 + 128 = 3072$.

SHA-512 Overall block diagram



Module F

Step1: message scheduling



$$W_t = \sigma_1^{512}(W_{t-2}) + W_{t-7} + \sigma_0^{512}(W_{t-15}) + W_{t-16}$$

Module F Step1: message scheduling

$$W_t = \sigma_1^{512}(W_{t-2}) + W_{t-7} + \sigma_0^{512}(W_{t-15}) + W_{t-16}$$

where

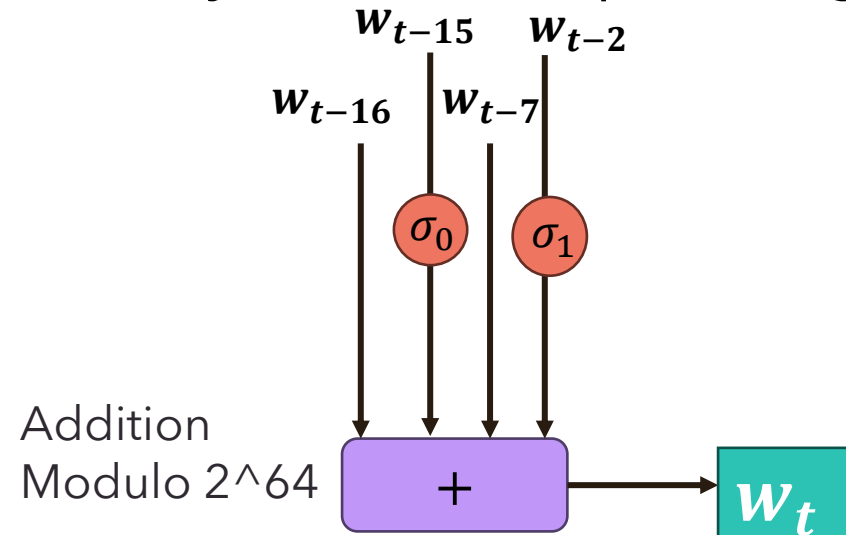
$$\sigma_0^{512}(x) = ROTR^1(x) + ROTR^8(x) + SHR^7(x)$$

$$\sigma_1^{512}(x) = ROTR^{19}(x) + ROTR^{61}(x) + SHR^6(x)$$

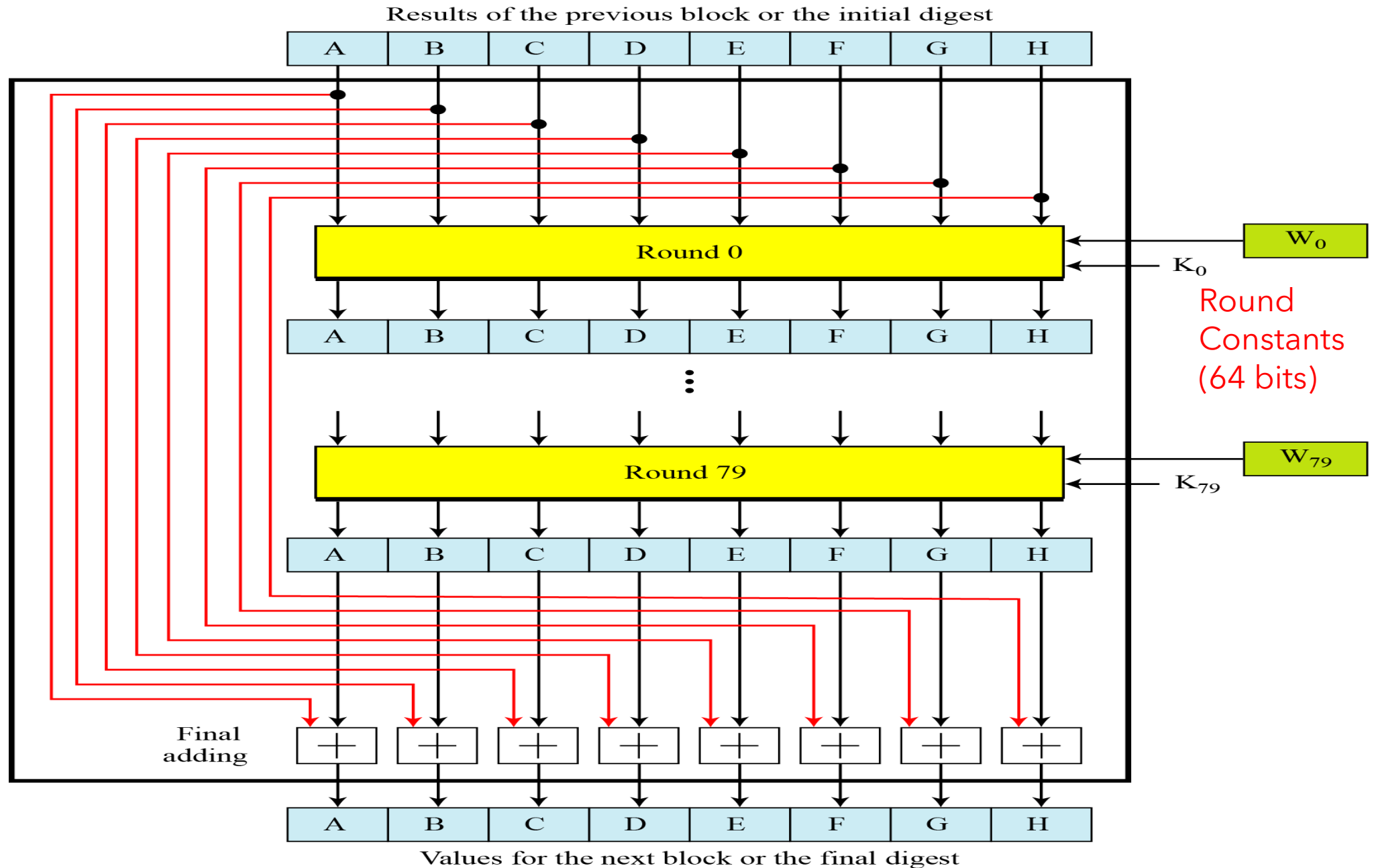
$ROTR^n(x)$ = circular right shift (rotation) of the 64-bit argument x by n bits.

$SHR^n(x)$ = left shift of the 64-bit argument x by n bits with padding by zeros on the right.

+ = addition modulo 2^{64}



SHA-512 Compression function



SHA-512 Initial Values & Word Expansion

Buffer	Value (in Hexadecimal)	Buffer	Value (in Hexadecimal)
A ₀	6A09E667F3BCC908	E ₀	510E527FADE682D1
B ₀	BB67AE8584CAA73B	F ₀	9B05688C2B3E6C1F
C ₀	3C6EF372FE94F82B	G ₀	1F83D9ABFB41BD6B
D ₀	A54FF53A5F1D36F1	H ₀	5BE0CD19137E2179

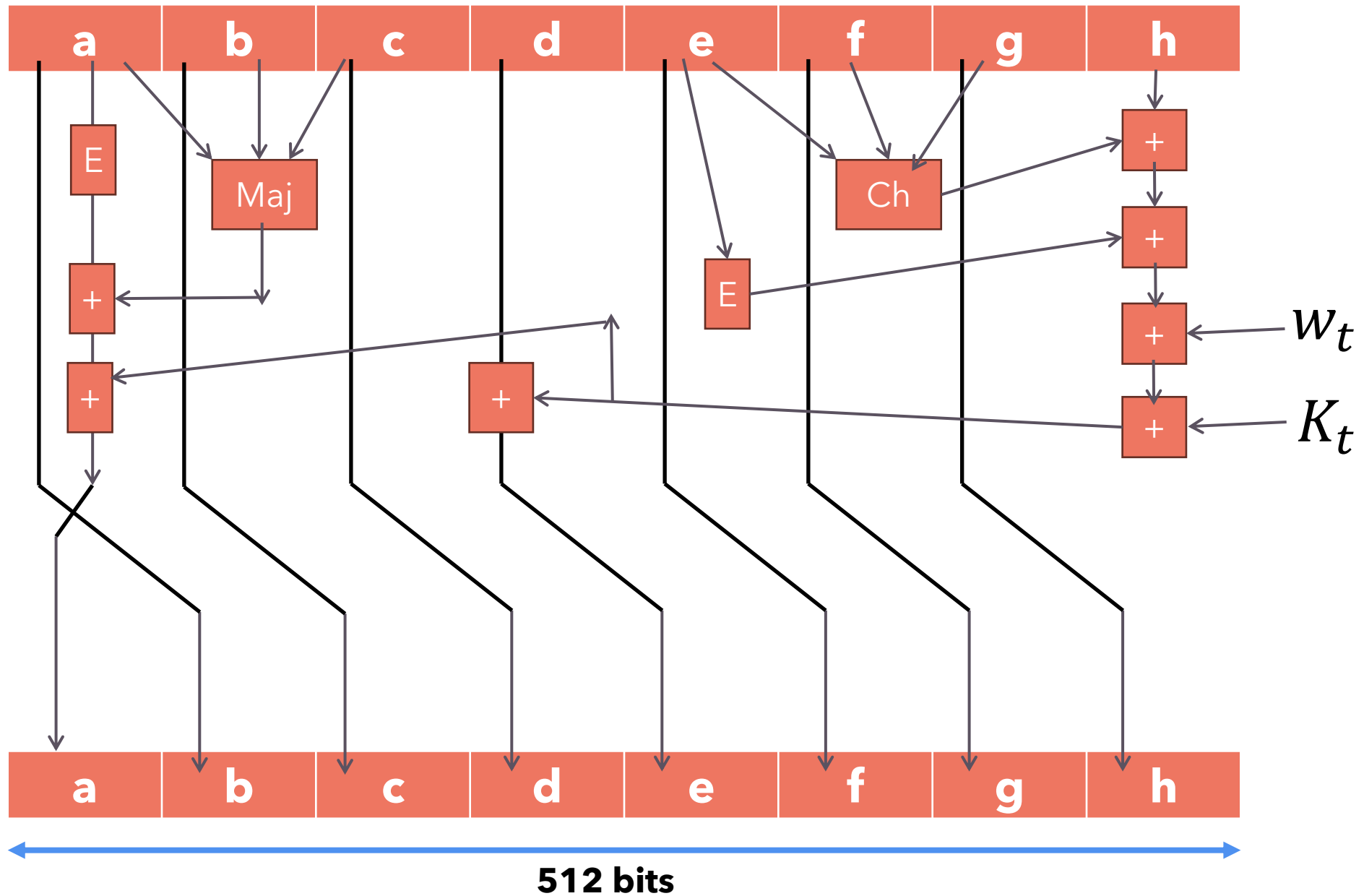
Table : Values of constants in message digest initialization of SHA-512

SHA-512 Round constants (K)

428A2F98D728AE22	7137449123EF65CD	B5C0FBCFEC4D3B2F	E9B5DBA58189DBBC
3956C25BF348B538	59F111F1B605D019	923F82A4AF194F9B	AB1C5ED5DA6D8118
D807AA98A3030242	12835B0145706FBE	243185BE4EE4B28C	550C7DC3D5FFB4E2
72BE5D74F27B896F	80DEB1FE3B1696B1	9BDC06A725C71235	C19BF174CF692694
E49B69C19EF14AD2	EFBE4786384F25E3	0FC19DC68B8CD5B5	240CA1CC77AC9C65
2DE92C6F592B0275	4A7484AA6EA6E483	5CB0A9DCBD41FBD4	76F988DA831153B5
983E5152EE66DFAB	A831C66D2DB43210	B00327C898FB213F	BF597FC7BEEF0EE4
C6E00BF33DA88FC2	D5A79147930AA725	06CA6351E003826F	142929670A0E6E70
27B70A8546D22FFC	2E1B21385C26C926	4D2C6DFC5AC42AED	53380D139D95B3DF
650A73548BAF63DE	766A0ABB3C77B2A8	81C2C92E47EDAEE6	92722C851482353B
A2BFE8A14CF10364	A81A664BBC423001	C24B8B70D0F89791	C76C51A30654BE30
D192E819D6EF5218	D69906245565A910	F40E35855771202A	106AA07032BBD1B8
19A4C116B8D2D0C8	1E376C085141AB53	2748774CDF8EEB99	34B0BCB5E19B48A8
391C0CB3C5C95A63	4ED8AA4AE3418ACB	5B9CCA4F7763E373	682E6FF3D6B2B8A3
748F82EE5DEFB2FC	78A5636F43172F60	84C87814A1F0AB72	8CC702081A6439EC
90BEFFFA23631E28	A4506CEBDE82BDE9	BEF9A3F7B2C67915	C67178F2E372532B
CA273ECEEA26619C	D186B8C721C0C207	EADA7DD6CDE0EB1E	F57D4F7FEE6ED178
06F067AA72176FBA	0A637DC5A2C898A6	113F9804BEF90DAE	1B710B35131C471B
28DB77F523047D84	32CAAB7B40C72493	3C9EBE0A15C9BEBC	431D67C49C100D4C
4CC5D4BECB3E42B6	4597F299CFC657E2	5FCB6FAB3AD6FAEC	6C44198C4A475817

Figure: List of round constants used in SHA-512

SHA-512 Round Function



T₁ and T₂ functions

$$T_1 = h + ch(e, f, g) + \Sigma_1^{512} e + W_t + K_t$$

$$ch(e, f, g) = e \text{ AND } f + (NOT\ e \text{ AND } g)$$

$$\Sigma_1^{512} e = ROTR^{14}(e) + ROTR^{18}(e) + ROTR^{41}(e)$$

$$ROTR^n(x)$$

= *Circular* right shift (rotate) of the 64
– bit argument x by n – bits

+ is XOR operation

T₁ and T₂ functions

$$T_2 = \Sigma_1^{512} a + \text{Maj}(a, b, c)$$

$$\text{Maj}(a, b, c) = (a \text{ AND } b) + (a \text{ AND } c) + (b \text{ AND } c)$$

$$\Sigma_1^{512} a = \text{ROTR}^{28}(a) + \text{ROTR}^{34}(a) + \text{ROTR}^{39}(a)$$

$$a = T1 + T2$$

$$h = g$$

$$e = d + T1$$

$$g = f$$

$$d = c$$

$$f = e$$

$$c = b$$

$$b = a$$